Research report: adding rehab data to sports 3D data lets single-camera 3D motion capture beat the Kinect on rehab movements

2026-10-31 · Sonny (AI application team) · status: exploratory, 3 seeds × 3 participant folds · 한국어: RESEARCH_REPORT_KR.md

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Conclusion (first page)

> **A 2D→3D model trained on sports data (ASPset-510) alone could not handle rehab movements. After post-hoc camera calibration of a rehab dataset (KERAAL) and mixing it into training, a single ordinary camera produced more accurate joint angles than a Kinect depth sensor on new participants never used in training.**

	ASPset only	ASPset + KERAAL	Kinect sensor
KERAAL joint-position error (MPJPE)	222.9 ± 3.0 mm	**69.5 ± 0.8 mm**	–
KERAAL joint-angle correlation (r0)	0.59	**0.76**	0.67
KERAAL joint-angle error (RMSE)	20.3°	**13.9°**	18.7°
Trunk-rotation exercise (RTK) angle correlation	0.23 (fails)	**0.72**	0.63
ASPset test MPJPE (original domain)	83.4 mm	**81.1 mm** (no loss)	–

Values are means ± SD over 3 seeds; KERAAL values are measured only on participants never used in training.

Five key findings 1. **The problem was pose variety, not data volume.** Going from 3 to 13 ASPset training subjects left the KERAAL error unchanged. 89% of KERAAL poses are seated, against 13% in ASPset. 2. **The calibration-free KERAAL dataset can be calibrated post hoc.** Using the Kinect as a bridge, Vicon was moved into camera coordinates: 25 mm 3D residual and 6.2 px reprojection error on held-out recordings. 3. **Merged training cuts position error to about a third.** 222.9 → 69.5 mm, better in all 9 seed × participant runs. 4. **It beats the Kinect on angles too.** This angle metric is independent of the calibration. r0 is higher and RMSE lower than the Kinect in all 9 cells (participant × exercise), p = 0.004. 5. **The RTK failure came from missing data, not a single-camera limit.** After merging, r0 rose from 0.23 to 0.72, past the Kinect (0.63).

Caveat: only 3 test participants (healthy, acted errors), filmed in the same room with the same camera as the training data. Real patients and other locations are not yet tested (§7).

1. Background and questions

- **Goal:** turn single-camera 2D video into 3D motion-capture data, for use in rehab exercise assessment.
- **Earlier results (10/26–10/30):**
- A model trained on ASPset-510 reached 49 mm PA-MPJPE within ASPset, on par with the paper's baselines.
- On KERAAL, however, angle RMSE was 20°, similar to the Kinect, and it failed on the RTK exercise.
- Training on 3/6/9/13 subjects left KERAAL performance unchanged.
- **Questions:**
- Q1. Can KERAAL get 3D ground truth in the same format as ASPset? (camera calibration)
- Q2. Does merged training help on new participants' rehab movements, and does it keep performance in the original domain (ASPset)?
- Q3. Does it matter whether Vicon or the Kinect is used as ground truth?

2. Data

	ASPset-510	KERAAL Group 3
Licence	CC0	CC BY-NC-SA (no raw or derived files in public)
Setting	outdoor, standing full-body sports	indoor, 3 seated arm rehab exercises (CTK, ELK, RTK)
People	17 (train 13 / val 2 / test 2)	3 (healthy, acting errors), 540 recordings
Video	4K 50 fps, 3 cameras	960×540 30 fps, 1 camera (Kinect v2 colour)
3D GT	markerless triangulation + bundle adjustment	Vicon optical markers at 60 fps, 17 rigid targets (position + quaternion)
Calibration	yes (§3.1)	**none → built by us (§3.2)**
Pose range (measured)	13% seated	89% seated; 62% of frames have no similar ASPset pose

3. Camera calibration methods

3.1 ASPset: the authors' method (Nibali et al. 2021, §4)

1. **Capture:** 3 × Canon EOS C200, 24 mm lenses, 4K 50 fps, about 30° apart around the stage. Synchronised by hand on a light being switched off.
2. **Initial calibration:**
3. An **asymmetric circle-grid board** was moved around the stage with twists and tilts.
4. **OpenCV** detected the circles to give K per camera, stereo calibration gave E, and gradient descent then reduced the reprojection error of all three cameras at once.
5. **Lens distortion was assumed zero.**
6. Detections whose circle rows/columns were not straight (MSE > 0.05) were dropped, and only frames at least 50 px apart were kept (~100).
7. **Bundle adjustment:**
8. Camera parameters were optimised jointly with 2D joint detections (Stacked Hourglass fine-tuned on human annotations).
9. It used 5,000 frames with score U < 10 px, which tolerates a constant offset across joints.
10. **Coordinates:** world = first (left) camera; mm; x right · y down · z forward. Each subject has its own calibration.
11. **Limitation:** no GT accuracy in mm is reported.

3.2 KERAAL: our post-hoc calibration (`keraal_calib.py`)

With no calibration board, we used the **Kinect as a bridge**. The structure is the same as ASPset: an initial estimate, then refinement with correspondences.

```
Vicon targets ──② fixed offsets──▶ Vicon joint points ──③ R, t──▶ camera coords ──① K──▶ pixels
                                                ▲
                                   Kinect joints (camera coords)
```

① **Intrinsics K** - Basis: - KERAAL video at 960×540 is exactly half of the Kinect v2 colour stream (1920×1080). - Kinect file rows match video frames 1:1. - Kinect joints are already stored in camera coordinates (m). - Method: - A pinhole model. From 12 joint correspondences between Kinect joints and YOLO 2D (confidence > 0.5), we fitted fx, fy, cx, cy together with a colour–depth sensor offset t0. - soft-L1 robust least squares. - Kinect axes (x left, y up) were flipped to OpenCV axes (x right, y down). - Result: **fx 502.3, fy 521.4, cx 476.0, cy 256.5, t0 = (4.8, −10.2, −19.6) cm**. Distortion assumed zero (same as ASPset).

② **Vicon targets → joint points** - The 17 Vicon items are rigid targets on body segments, not joint centres. Matching them directly to the Kinect left about 12 cm of residual. - So joint point = target position + target rotation × **fixed offset**, where the offset is a constant in the target's local frame.

③ **Vicon → camera rigid transform** - Kinect joint ≈ R·(joint point) + t, with scale fixed in mm. - R, t (Kabsch) and the offsets (closed-form median) were solved by **alternating least squares**, 30 iterations. Halfway through, frames with Kinect tracking failures (residual above 3× the median) were dropped once. - Vicon was decimated from 60 to 30 fps, and a time lag of ±15 frames was searched per recording. - **Session = participant × exercise.** With one session per participant, only P2 failed (111 mm); per exercise it gave 25–33 mm. - Output: Vicon and Kinect joints per recording, **in ASPset `aspset_17j` order and camera mm**. head_top has no counterpart and is left empty.

Calibration check (fitted on even-numbered recordings, checked on the 270 odd-numbered ones)

Check	Median
Vicon joint points ↔ Kinect joints (3D)	**25.0 mm** (P1 22, P2 29, P3 23)
Vicon projection ↔ YOLO 2D	**6.2 px** (5.9 px with the ASPset-style U score)
Kinect projection ↔ YOLO 2D	6.8 px
Offset sizes	mostly 80–230 mm
Visual check	Vicon points sit on the body's joints in held-out frames

4. Experiment design

Item	Setting
Input	YOLO11x-pose 2D (COCO 17) + confidence, mid-hip + body-extent normalisation
Model	temporal conv net (VideoPose3D-style), 81-frame receptive field, output pelvis-relative 3D for 17 joints
Split	**leave-one-KERAAL-participant-out, 3 folds.** 1/5 of the training participants' recordings held out as KERAAL validation. The test participant is used only at the end
Model selection	epoch with the lowest (ASPset val + KERAAL val MPJPE) / 2
Loss	per-joint masked MPJPE; joints without GT are left out
Conditions	① ASPset only ② ASPset + KERAAL (Vicon GT) ③ ASPset + KERAAL (Kinect GT) ④ KERAAL only (Vicon GT, 457 epochs for the same number of updates)

Item	Setting
Training	AdamW lr 1e-3, 120 epochs, left-right flip augmentation + flip TTA, **seeds 0, 1, 2**
Metrics	**Position:** MPJPE and PA-MPJPE against Vicon joint points (16 joints, no head_top). **Angles:** correlation r0 and RMSE against Vicon **quaternion** relative angles; the angle metric is independent of the calibration. **ASPset test:** checks the original domain is kept
Tests	Wilcoxon signed-rank over 9 test-participant × exercise cells (seed-averaged)

5. Results

5.1 Main results (mean ± SD over 3 seeds, averaged over the 3 participant folds)

Training data	KERAAL MPJPE	PA-MPJPE	Angle r0	Angle RMSE (°)	RTK r0	ASPset MPJPE / PA
① ASPset only	222.9 ± 3.0	102.9 ± 0.1	0.594 ± 0.017	20.3 ± 1.4	0.233	83.4 / 49.4
② **ASPset + KERAAL (Vicon)**	**69.5 ± 0.8**	**52.2 ± 0.2**	**0.764 ± 0.005**	**13.9 ± 0.3**	**0.724**	**81.1 / 48.5**
③ ASPset + KERAAL (Kinect)	77.7 ± 0.2	56.9 ± 0.2	0.712 ± 0.005	15.7 ± 0.4	0.616	81.9 / 48.8
④ KERAAL only (Vicon)	68.4 ± 0.5	52.8 ± 0.5	0.746 ± 0.004	13.6 ± 0.1	0.739	315.4 / 212.1
Kinect sensor	–	–	*0.670*	*18.7*	*0.634*	–
Raw 2D angles	–	–	*0.551*	*29.1*	–	–

5.2 Per participant (seed-averaged)

Training data	P1 MPJPE / r0	P2 MPJPE / r0	P3 MPJPE / r0
ASPset only	189.5 / 0.78	234.5 / 0.59	244.5 / 0.41
ASPset + KERAAL (Vicon)	82.4 / 0.88	70.2 / 0.72	55.9 / 0.69
KERAAL only	92.5 / 0.84	60.3 / 0.72	52.3 / 0.68

5.3 Position MPJPE per exercise (mm)

Training data	CTK	ELK	RTK
ASPset only	230.6	208.7	228.8
ASPset + KERAAL (Vicon)	70.3	61.1	77.5
ASPset + KERAAL (Kinect)	74.4	74.0	85.0
KERAAL only	70.7	58.3	76.3

5.4 Per-joint MPJPE (mm, ASPset only → ASPset + KERAAL Vicon)

Joint	Before → after	Joint	Before → after
Ankle R / L	431 → **58** / 434 → **65**	Wrist R / L	298 → 147 / 287 → 150
Knee R / L	376 → **55** / 360 → **75**	Elbow R / L	212 → 102 / 212 → 104
Hip R / L	83 → 12 / 83 → 11	Shoulder R / L	169 → 81 / 174 → 74
Head / neck / spine	154 → 70 / 170 → 67 / 120 → 44		

5.5 Statistical tests (9 cells = participant × exercise, seed-averaged)

Comparison	r0 diff, cells better, p	RMSE diff, cells better, p
② vs ① ASPset only	+0.256, **9/9**, p = 0.004	−6.1°, 8/9, p = 0.012
② vs Kinect	+0.103, **9/9**, p = 0.004	−7.9°, **9/9**, p = 0.004
③ vs Kinect	+0.041, 6/9, p = 0.129	−6.0°, 9/9, p = 0.004
④ vs Kinect	+0.088, 8/9, p = 0.020	−8.1°, 9/9, p = 0.004
② Vicon GT vs ③ Kinect GT	+0.062, 7/9, p = 0.039	−1.8°, 7/9, p = 0.055
② merged vs ④ KERAAL only	+0.015, 5/9, p = 0.43	+0.3°, 2/9, p = 0.16

In all 9 seed × participant runs, ②, ③ and ④ had lower position error than ①.

6. Interpretation: why it worked

1. **It filled the missing poses.** The lower body improved most: ankle 431 → 58 mm, knee 376 → 55 mm. The ASPset-only model predicted seated legs as if standing, which matches the measured pose range (13% vs 89% seated).
2. **The RTK failure was a data problem.** r0 rose from 0.23 to 0.72. The earlier reading "rotation in depth = single-camera limit" was wrong.
3. **The GT format matched ASPset exactly.** The §3.2 calibration put KERAAL GT into the same frame (OpenCV axes, mm, pelvis-relative, same 17-joint order), so the data could be mixed without changing the model or the loss.
4. **Calibration error is below model error.** Calibration 25 mm vs model 69 mm, so GT noise did not spoil training.
5. **The two datasets protect each other.**
6. On KERAAL, ② and ④ are statistically the same (p = 0.43).
7. But ④ collapsed on ASPset (315 mm), while ② even improved on ASPset (83.4 → 81.1 mm).
8. So **one model for both domains needs the merge**. The paper's §6.2 (MPI-INF-3DHP + ASPset) shows the same effect.
9. **GT quality matters.** Vicon GT (②) beat Kinect GT (③) even on the calibration-independent angle r0 (p = 0.039), so §3.2 steps ②③ (moving Vicon into the camera frame) were worth doing.
10. **The results are stable.** Between-seed SD is 0.8 mm for MPJPE and 0.005 for r0, and all 9 seed × participant runs improved.

7. Limitations, failures, and why

1. **P2 failed in the first calibration (111 mm).**
2. P2's camera or Vicon frame apparently changed between exercises; per-exercise calibration fixed it.
3. Whether the camera actually moved is unchecked. P2 also has many Vicon dropouts (42 recordings over 20% missing).
4. **The ASPset-only position error (223 mm) includes a joint-definition mismatch.**
5. KERAAL GT follows the Kinect definition because the offsets were fitted to Kinect joints; ASPset follows anatomical annotation.
6. So part of the position gain is "learning the definition". The angle metric is definition-free and also improved a lot, so the conclusion holds.
7. **Wrist error is still large (~150 mm).** Wrist GT is a forearm-target offset fitted to the Kinect wrist, whose tracking is unstable. Functional joint centres (SCoRE) or the hand targets could fix this.

8. **A KERAAL-only model does not generalise.** Two people, indoors, three exercises — too narrow a pose range.
9. **The sample is small.** 3 test participants; the 9 cells are three exercises from the same person each, so they are not fully independent.
10. **Same room, same camera.** We showed generalisation to new people, not to new places or cameras.
11. **Only acted errors were tested.** Real patients (1A) may move differently.
12. **ASPset's own GT accuracy is unreported.** The paper gives no validation in mm.

8. Next steps

Order	Task	Purpose
1	Apply the merged model to 1A (patients) and 2A (healthy) videos and compare angles with the Kinect	generalisation to real patients and other sessions (no Vicon → angles only)
2	Rebuild GT with functional joint centres	better wrists, no dependence on the Kinect definition
3	Rerun error detection (RQ4) with the merged model	link to the application goal
4	External check on OpenCap	different place and camera
5	Joint centres + IK → BVH	complete the "motion-capture data" goal

9. Reproduce

```
python keraal_calib.py --check
python keraal_calib.py E:/KERAAL_data/extracted/group3 E:/KERAAL_mocap3d/det2d E:/KERAAL_mocap3d/calib
python merge_train.py --check
python merge_train.py <ASPset root> <ASPset det> <G3> <G3 det> E:/KERAAL_mocap3d/calib <run> --hold P1 --
mode aspset_vicon --epochs 120 --seed 0
python merge_train.py ... <run> --hold P1 --seed 0 --eval-only E:/ASPset510_data/runs/det_root_s0   # ①
ASPset only
python keraal_eval.py eval <G3> <G3 det> <run> <run>/angles.json
```

- Environment: `E:\mocap3d_env` (Python 3.12, PyTorch, ultralytics), RTX 3090.
- Results: `E:\KERAAL_mocap3d\merge\summary_3seeds.json`, per fold/seed `results.json` and `angles.json`.
- Calibration: `E:\KERAAL_mocap3d\calib\calib.json`.
- **Anything derived from KERAAL (GT, figures) is CC BY-NC-SA and must not go on GitHub.**
- Earlier records: `RESULTS.md` (baseline), `MERGE_REPORT_EN.md` (single-seed first report).

References

- Nibali, Millward, He, Morgan. ASPset: An outdoor sports pose video dataset with 3D keypoint annotations. *Image and Vision Computing* 111 (2021) 104196.
- Nguyen et al. KERAAL dataset. arXiv 2407.00521.
- Pavllo et al. 3D human pose estimation in video with temporal convolutions and semi-supervised training (VideoPose3D). CVPR 2019.